

PSO-Based Fuzzy Logic Optimization of Dual Performance Characteristic Indices for Fast Charging of Lithium-Ion Batteries

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Abstract—Efficacies of the charging strategy and electrified pattern dominate the functionality and lifespan of Lithium-ion (Li-ion) batteries intensely. In order to maximize the available performance of the Li-ion batteries, in this paper, a searching strategy based on the particle swarm optimization (PSO) conducted with a fuzzy-deduced fitness evaluator (FDFE) is proposed to find the best multistage charging current pattern. The objective function is to maximize the cost benefit of the applied prospective charging pattern based on the nonlinear weightings' allocation of the charge time (CT) and normalized discharge capacity (NDC) in the objective function. Rules of the weighting allocation are derived from the fuzzy logic inference. The fitness value of each candidate particle (charging pattern) is figured out by the FDFE to guide the searching path of the PSO algorithm for finding the optimal solution out. Experimental findings show that the resulted pattern is capable of charging the batteries to over 90% available capacity within 50 minutes. Comparing with the conventional constant current–constant voltage (CC-CV) method, the devised scheme has the performance enhancements of more than 80% charging time reduction, 21% more life cycles, and over 0.4% charging efficiency increase.

I. INTRODUCTION

Rechargeable batteries play a significant role in energy storage solutions for the state-of-the-art battery-powered technologies. Proliferation of portable mobile communication devices leads to the demand for rechargeable batteries with different capacities and cell chemistries increasingly and a considerable impact on the environment. Among the commonly used rechargeable batteries, the lithium-ion (Li-ion) batteries are a popular choice for applications in modern electronic appliances [1]–[4], such as hand-held consumer electronics, aerospace, transportation vehicles, and energy backup for renewable energy systems etc., by reason of their attractive characteristics like more environmentally friendly, high energy and power density, high open-circuit voltage (OCV), low self-discharge rate, less maintenance, and immunization of memory effect.

Fast charging has always been an essential design consideration for the battery-powered device. Therefore, it has been becoming a focus on charger developments as how to make fast charge achievable, yet still maintain high efficacy. The CC-CV method, however, is unsuitable for fast charging because of the CV charging phase significantly prolongs the charging time and also degrades the energy efficiency. To cope with this problem, several researches have been devoted to the study of the charging techniques and topologies. A phase-shift full-bridge converter is proposed in [5] to be the Li-ion battery charger to promise higher output capability and conversion efficiency. This charger is controlled by a microcontroller which is responsible for steering it to meet CC-CV charging profile. In [6], a **smooth control circuit with a built-in resistance compensator method is presented to switch the mode transition from CC to CV stage smoothly and redistribute the operating durations of the two dominant phases to speed up the charging process effectively.** References [7] and [8] address a series of improvements in charging strategies, waveforms, or circuit topologies to satisfy fast charging requirement. These ameliorations and schemes indeed make better improvement in charging efficiencies and charging speed. On the other hand, the advantages of the multistage CC-charging methods [9], such as less chemical reaction stress and shorter charging time, have been confirmed. However, the charging pattern was only obtained from the electrochemical behavior of batteries, which was not promised to be the best solution. Recently, the intelligent control algorithms, such as fuzzy, neural network, grey, and genetic approaches [10]–[12], have been integrated into the commercial microcontrollers or customized ICs to develop chargers, in order to improve the accuracy of the parameters' prediction and operating performance of the chargers further. In spite of these solutions have made great progress in charger design, the accurate operating traits and mathematical model of the battery under charge generally are necessary for the algorithms as the analysis and computation basis. However, a precise model of Li-ion battery is very difficult to obtain due to the nature of the complicated electrochemical properties. Therefore, the

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improvement only in charging modes could not promise the best charging profiles and significantly boost the charging speed and efficacy. Furthermore, battery cycle life issues are seldom discussed in these researches.

In this paper, a global optimization technique based on the PSO algorithm is introduced to fast charging-pattern searching problems for enhancing the available performance of the Li-ion batteries. On the basis of the decision-making of the control process, the fitness value of the prospective charging pattern within the solution space is measured to direct the PSO search to find out the optimal solution. Two paramount parameters, i.e. charge time and discharge capacity, are employed to formulate the objective (cost) function of the charging policy. Allocating proportions of the weightings about the charge time and discharge capacity within the cost function are originated from the fuzzy logic inference to maximize the cost effectiveness. **The proposed optimal searching method** features fast convergence, effectiveness, and easy to implement. The implementation procedure will be described in detail, and the design considerations and realization example are presented to demonstrate its workability and validity. Experimental results demonstrate that the optimal charging pattern obtained by the presented algorithm can make the desired performances, such as more discharge capacity, shorter charging time and longer life cycle, achievable.

II. PSO-BASED FAST CHARGING PATTERN SEARCH

A. Fundamentals of PSO Operation

The PSO is a population-based evolutionary computational technique, which was inspired by the observation of the collective foraging social behavior in the bird flocks or fish schools, developed by Eberhart and Kennedy first in 1995 [13]. The PSO algorithm can serve as a powerful optimizer to deal with global optimization problems having single- or multi-objective functions with several local optimum values. The most remarkable property of the PSO approach is its simplicity and effectiveness because only two model equations are involved. In order to improve the poor convergence characteristic, several variants of the PSO algorithm are evolved, yet the most popular is the global formulation (G_{best} model) devised by Eberhart and Shi.

Compositions dominating the flying experience of each particle in its search for optimal solution are shown in Fig. 1. The movement of each particle is dragged by two tractions in accordance with two simple formulae mainly, one drawing it with random magnitude to the fittest location so far encountered by the particle, and one attracting it with magnitude at random to the best place met by any of the particle's social neighbors in the swarm. Namely, particles moving are guided by their own known best position in the search space as well as the entire swarm's best known position. When present best positions are being discovered, these will then come to conduct the movements of the swarm. Then, each particle updates its searching trace based on its own best local exploration experience (P_{best}) and the best global one (G_{best}) according to the iterative model equations as follows:

$$v_i^n(k+1) = \underbrace{wv_i^n(k)}_{\text{previous velocity}} + \underbrace{\Lambda_1 \mathfrak{R}_1 (P_{best,i}^n - x_i^n(k))}_{\text{cognitive component}} + \underbrace{\Lambda_2 \mathfrak{R}_2 (G_{best}^n - x_i^n(k))}_{\text{social component}} \quad (1)$$

$$x_i^n(k+1) = x_i^n(k) + v_i^n(k+1) \quad (2)$$

Where $x_i^n(k+1)$ and $x_i^n(k)$ represent the current and previous positions in the n th dimension of particle i , respectively; $v_i^n(k+1)$ and $v_i^n(k)$ stand for the current and previous velocities, respectively. The w is an inertia weight. The accelerative constants Λ_1 and Λ_2 are the cognitive and social learning factors, respectively. The $\mathfrak{R}_1$ and $\mathfrak{R}_2$ are two random values in the range of $[0, 1]$. G_{best}^n is the current location with the best fitness. $P_{best,i}^n$ is the individual location had the best fitness measure, and k is the iteration index.

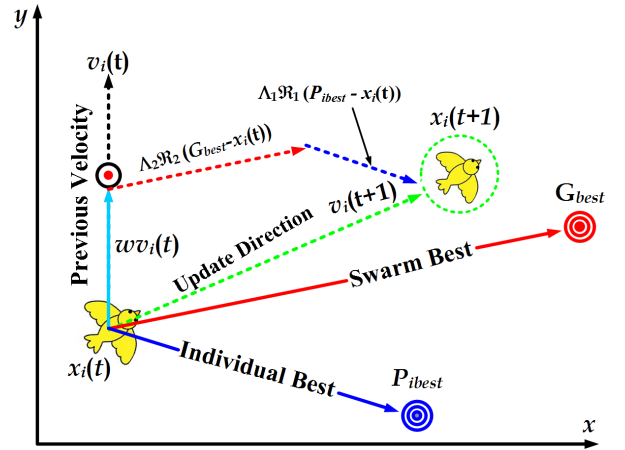


Figure 1. Concept of update in a searching particle by PSO.

B. Five-Stage Constant-Current Charge

Short charging time, extendable cycling life and efficient energy transfer are the major measures of a high-quality fast charger. The multistage CC charging technique has been confirmed that it has the ability to fulfill the fast-charge criteria. It can particularly mitigate the chemical reaction stress and temperature rise of the batteries under charge. Therefore, in this paper, the five-stage CC charging approach is utilized to perform rapid charge for the Li-ion batteries. Fig. 2 illustrates the basic concept of the multistage CC charging method. The total charging period, as shown in Fig. 2, is divided into five stages. The charging current in each stage is preset to a default value. During the charging phase, the battery voltage will increase gradually. The charge process will enter the next stage and a new set of charging current will be applied accordingly as soon as the voltage arrives at the default limit voltage (V_{lim}). This process will continue until the charging stage number reaches 5. The gradual reduction mechanism of the charging current cooperating with appropriate rest time makes the multistage CC method easier to balance off the chemical reaction concentration and subdue the temperature rise of the battery, and resulting in shorter charging time and more running cycles.

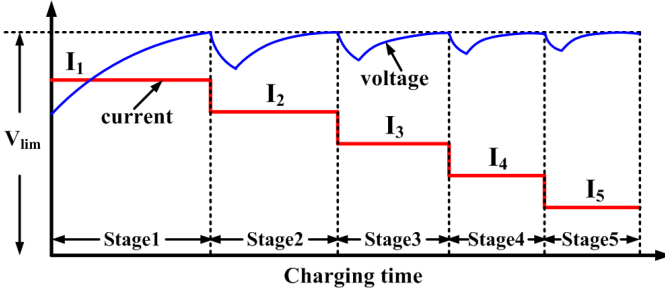


Figure 2. Basic structure of the multistage CC charging method.

C. Problem Formulation

Precise behavior of the Li-ion batteries is not easy to model mathematically due to their complicated electrochemical characteristics. Accordingly, it is also hard to determine the optimal current level for each charging stage only by using battery models. Through traditional skills to solve fast charging problem economically is difficulty and often expensive because most of conventional fast charges lean on a critical-point based approach. Doing in this manner although can protect the cells from undue thermal and chemical stresses, it reduces the batteries life expectancy as well. In addition, the available life of a rechargeable battery not only depends on the charging time, but also on the charging pattern and overcharging or undercharging control. This paper aims at determining the best pattern of the five-stage charging currents to meet the rapid-charge criteria (the fully charged battery must have the maximum discharge capacity within the shortest charging time) and achieving the best charging cost benefit. Therefore, an optimal charging pattern search strategy using PSO algorithm is proposed to work the oracle, and the objective function of this problem is formulated as

$$\begin{aligned}
 & \text{Maximize } f_{ce}^i(T_{ch}^i(\vec{I}_j^i), C_{dis-norm}^i(I_{dis}^i)), \quad \vec{I}_j^i \in \mathbb{S} \\
 & \text{Where } \vec{I}_j^i = [I_1^i, I_2^i, I_3^i, I_4^i, I_5^i] \\
 & \text{Subject to } 30 \text{ min} \leq T_{ch}^i(\vec{I}_j^i) \leq 90 \text{ min} \\
 & \quad 80\% \leq C_{dis-norm}^i(I_{dis}^i) \leq 100\% \\
 & \quad I_j^i \geq I_m^i \quad \text{if } j < m, \quad j, m = 1, 2, \dots, 5
 \end{aligned} \tag{3}$$

For battery i , the objective is to maximize the cost-effectiveness function (CEF) f_{ce}^i , which consists of dual performance traits index (DPTI), i.e. charging time (CT) T_{ch}^i and normalized discharge capacity (NDC) $C_{dis-norm}^i$. In which the CT is chiefly dominated by the five-stage charging current levels, $\vec{I}_j^i$, while the NDC is mainly related to the discharging current I_{dis}^i . The NDC is defined as the percentage ratio of the discharge capacity obtained by the PSO-based five-stage charge to that of the CC-CV method, both with the same discharge rate (0.1C) during the discharge phase to justly estimate the output capacities. Therefore, the NDC can be expressed as

$$C_{dis-norm}^i = \frac{C_{dis-5-stage}^i}{C_{dis-CC-CV}^i} \tag{4}$$

Where $C_{dis-5-stage}^i$ and $C_{dis-CC-CV}^i$ are the discharged capacities obtained from the PSO-based five-stage charging and the CC-CV charging techniques, respectively. $\mathbb{S}$ stands for the set of all feasible charging patterns. The constraints on the CT and NDC, in this paper, are specified to the range [30, 90] min and [80, 100] %, respectively, to meet the commonly considerations of the fast charging criterion. On the other hand, it should be noted that the charge level of the present running stage should not be greater than that of the preceding stage. Owing to the voltage limit value of each stage is the same, if the charging C rate of the current stage is greater than that of the previous stage, the battery terminal voltage will arrive at the voltage limit at once and the duration of this stage will be very short.

D. Fuzzy-Deduced Fitness Evaluator (FDFE)

Fitness value in the PSO-based optimization problem is used to evaluate the profit and performance of the applied charge pattern for each battery in the swarm for inspecting whether the cost effectiveness is the best or not. Any one of the particles which has the best fitness value will attract and guide the other particles in the swarm to follow its searching trajectory for achieving their correspondent objective. As a result, in order to find the authentic leading particle out, the fitness value of each particle must be measured exactly. According to the definition in (3), the cost effectiveness is a function of the charge time and discharge capacity that are added with particular weighting factor to stress their contributions to the CEF. Hence, on the studied issue, the fitness function $F^i(\cdot)$ for the battery i can be expressed as

$$F^i(T_{ch}^i, C_{dis-norm}^i) = f_{\alpha}^i(T_{ch}^i)T_{ch}^i(\vec{I}_j^i) + f_{\beta}^i(C_{dis-norm}^i)C_{dis-norm}^i(I_{dis}^i) \tag{5}$$

Where $f_{\alpha}^i(\cdot)$ is the nonlinear function of weighting α with respect to the quantity of the charging time, and $f_{\beta}^i(\cdot)$ is the nonlinear function of weighting β corresponding to the quantity of the normalized discharge capacity.

However, according to the understanding of the battery's electrochemical characteristics and the field-tested experience, we know that the charging time and discharge capacity are strongly depend on the charging/discharging current levels and they are conflictive each other nearly. Therefore, how to properly allocate the weighting factors $f_{\alpha}^i(\cdot)$ and $f_{\beta}^i(\cdot)$ in (5) to obtain the maximum charging cost effectiveness is significantly critical here. Obviously, constant fifty-percent-ratio weighting value dispatch for $f_{\alpha}^i(\cdot)$ and $f_{\beta}^i(\cdot)$ could not be promised the best fitness can be attained. Accordingly, a fuzzy-deduced fitness evaluator (FDFE) is proposed in this paper. The optimal regulating rules of weighting factors are formulated by the fuzzy logic (FL) inference to carry out the best fitness value evaluation. Shown in Fig. 3 is the scheme of the proposed FDFE for the fitness value assessment. Form Fig. 3, the five-stage charging time and discharge capacity

percentage are used as the input parameters, and the output parameter is evaluative value of the fitness function. The Mamdani-type minimum inferential method cooperating with the center-of-area (COA) defuzzification procedure is utilized to produce the crisp controller output. The membership functions (MFs) corresponding to the charging time, normalized discharge capacity, and fitness evaluation value are illustrated in Fig. 4(a)-(c), respectively. Here in all three MFs, except for the two fuzzy sets at the outmost ends (trapezoidal MFs are considered), symmetric triangles with equal base and fifty-percent overlap with adjacent MFs are chosen. The universe of discourse (UOD) in the MFs of the input variables, T_{ch}^i and $C_{dis-norm}^i$, are defined on the domain [30, 90] min. and [80%, 100%], respectively, to agree with the constraints specified in the problem formulation. In addition, the UOD in the MF of the output variable, ΔF_N^i , is defined on the normalized domain [0, 1].

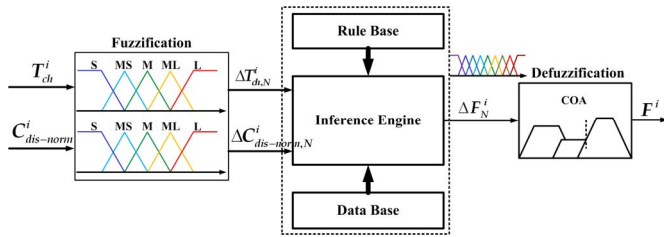


Figure 3. Scheme of the proposed FDFE.

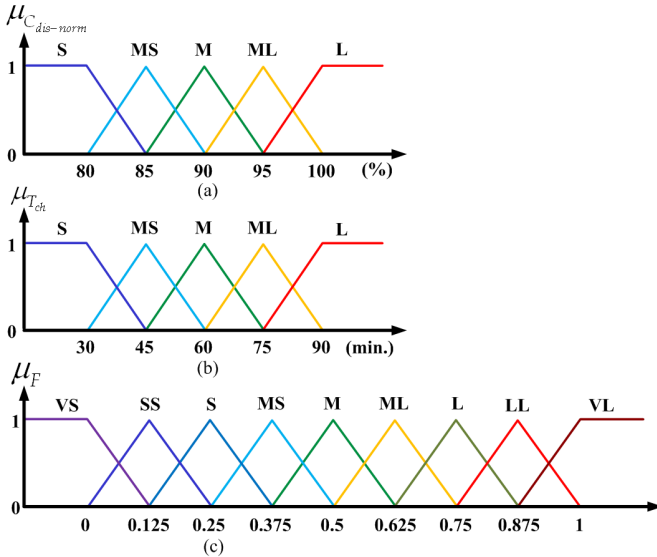


Figure 4. MFs of the input & output variables.

The linguistic value S denotes short or small in charging time or discharge capacity and L represents long or large in charging time or discharge capacity respectively. Such that the VS, SS, MS, M, ML, LL, and VL stand for very small, small small, medium small, medium, medium large, large large, and very large, respectively. The FL controller structure approximates the way of expert operation intuitiveness; this makes it attractive and easy to incorporate heuristic rules that reflect the experience of human experts into the controller. The rule bases for deducing the fitness value is shown in Table I. This is a widely used rule base designed with a two-

dimensional phase plane. As shown in Table I, the control rules are reasoned to obtain the optimal cost effectiveness based on the perception that the larger discharge capacity and the shorter charging time will obtain the higher cost benefit, and on the contrary, the smaller discharge capacity and the longer charging time will result in the lower cost benefit. For example, if the discharge capacity is large and the charging time is short, then the fitness evaluation is very large. The rule can be formed as: If $C_{dis-norm}^i$ is L and T_{ch}^i is S, then ΔF_N^i is VL.

TABLE I. RULE BASED DERIVATUON FOR FITNESS EVALUATION

$C_{dis-norm}^i \backslash T_{ch}^i$	S	MS	M	ML	L
S	M Rule1	ML Rule2	L Rule3	LL Rule4	VL Rule5
MS	MS Rule6	M Rule7	ML Rule8	L Rule9	LL Rule10
M	S Rule11	MS Rule12	M Rule13	ML Rule14	L Rule15
ML	SS Rule16	S Rule17	MS Rule18	M Rule19	ML Rule20
L	VS Rule21	SS Rule22	S Rule23	MS Rule24	M Rule25

E. PSO-Based Search Procedure

Shown in Fig. 5 is the operating flowchart of the PSO algorithm applied in this paper. The main operating procedures are described as follows:

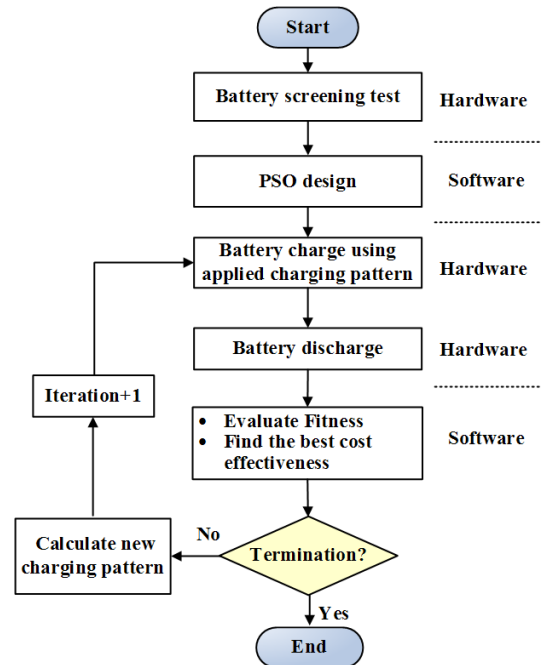


Figure 5. Flowchart of the proposed method.

Step1) Battery screening test: All batteries had to pass screening tests with acceptable open-circuit voltage (OCV), impedance, weight, and rated capacity values.

Step2) System initialized and PSO parameter design: Initialize the searching space according to the number of stages and charging current candidates and choose the parameters for PSO. In this paper, the initial swarm is first randomly generated. A random number generator is programmed to generate the initial five-stage charging current patterns of the batteries within the feasible search space. The charging pattern of each battery is also outputted to a file for subsequence use.

Step3) Charging/discharge batteries: The charging process was carried out by means of a WonATech WBCS3000 programmable charge/discharge tester. When the charging process is completed, the discharging processes are also conducted using the same equipment. The obtained discharge capacity and charging time for each battery are recorded in an output file for the fitness evaluation.

Step4) Fitness evaluation: According to the obtained NDC and CT, the FDFE computes the fitness value of each battery in the swarm.

Step5) P_{best} and G_{best} update: For each time search, comparing each battery's fitness with its previous best fitness (P_{best}) obtained. If the current value is better than old P_{best} , then update P_{best} . The next compares P_{best} of all batteries with each other and update the swarm global best location with the greatest fitness (G_{best}).

Step6) Generate new charging pattern: If the termination condition is unreached, the incremental value of each stage charging current for each battery will be changed towards its P_{best} and G_{best} . This acceleration is weighted by a random term. Then, a new charging pattern is generated by adding the new incremental current value to the previous charging current according to (1) and (2).

Step7) Repeat steps 3 to 5 until convergence is reached based on the two criteria: a) the difference of the average fitness value between the before and after iteration is less than 0.005; b) the standard deviation (STDEV) less than 0.01.

III. EXPERIMENTAL RESULTS

Fig. 6 shows the configuration of the experimental system. The battery test equipment used in this paper is WBCS3000 (5A, 5V, 8 channel) from WonATech corporation. In the screen stage, sixteen batteries are selected for experiments from fifty commercially available Li-ion batteries. Therefore, before the optimum search process, all the batteries tested in this paper have similar electrochemical characteristics and passed screening tests. From Fig. 6, the right-hand side of the dotted line is the software part, where the PSO algorithm is implemented. The left-hand side of the dotted line is the hardware part, which primarily is a battery tester with programmable ability. The relative parameters in the PSO method are: the w is 0.1; Λ_1 and Λ_2 are equal to 2; $\mathfrak{R}_1$ and $\mathfrak{R}_2$ are random numbers between 0 and 1. The specification of the test battery is 3.6V, 2200 mAh.

Table II tabulates the test results from the first one to the seventh test. The fitness value (fuzzy output), P_{best} , G_{best} , and the five-stage current setting values are recorded. According to the (1) and (2), the incremental current value and new charge current pattern are generated. This process is continuous until the convergence criteria are met. From Table II, at the seventh test, the difference of the average fitness value between the sixth and seventh iteration is about 0.005, it is less than 0.05; and the standard deviation (STDEV=0.007089) has been less than 0.01. They have agreed with the specified convergence criteria. Therefore, stop the searching procedure. Moreover, from the tabulated records in Table II, we can see that the proposed PSO-based method features fast convergence performance and battery #7 is the leading particle because it almost possesses the best fitness value than those of the other particles in each searching process. Shown in Fig. 7-9 are the convergence curves corresponding to the fitness value (fuzzy output), CT, and NDC, respectively. From Fig. 7, the other batteries' fitness values follow the leading battery's fitness values to converge toward the global maximum. In other words, according to the experiential communications of the individual cognitive share and global social share, the new incremental current values and charge current patterns of the batteries will iteratively update and follow that of the leading battery during the searching process, and finally the best charge current pattern will be obtained when the convergence criteria are fulfilled. Similarly, the behavior on the CT and NCD curves have the same convergence characteristics, i.e. a battery charged by the best charging pattern it will acquire the most charge capacity within the shortest charging time.

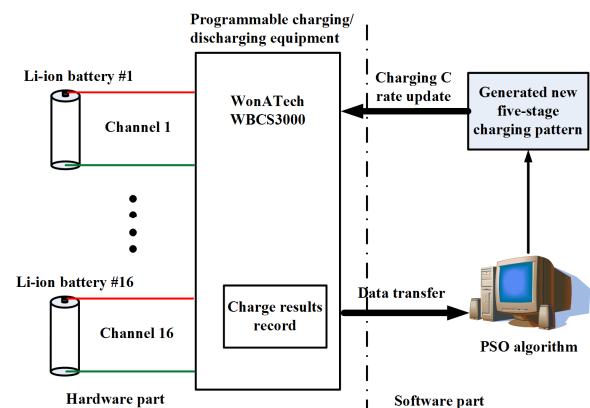


Figure 6. Experimental system setup.

TABLE II. TEST RESULT TABULATION

Battery no.	Fuzzy Output7	Fuzzy Output6	Fuzzy Output5	Fuzzy Output4	Fuzzy Output3	Fuzzy Output2	Fuzzy Output1	Pbest	Gbest	Battery no.	I ₁	I ₂	I ₃	I ₄	I ₅
B1	0.586	0.586	0.559	0.537	0.528	0.551	0.516	0.586	B1	1.43	1.12	0.87	0.73	0.39	
B2	0.581	0.566	0.558	0.543	0.531	0.557	0.517	0.581	B2	1.43	1.18	0.86	0.71	0.40	
B3	0.593	0.567	0.584	0.546	0.545	0.572	0.417	0.593	B3	1.44	1.33	0.88	0.69	0.38	
B4	0.583	0.597	0.557	0.553	0.561	0.548	0.502	0.597	B4	1.50	1.16	0.86	0.74	0.41	
B5	0.582	0.591	0.592	0.554	0.559	0.501	0.515	0.592	B5	1.47	1.17	0.88	0.72	0.41	
B6	0.577	0.566	0.543	0.541	0.546	0.557	0.556	0.577	B6	1.45	1.19	0.79	0.71	0.40	
B7	0.594	0.598	0.592	0.564	0.569	0.577	0.596	0.598, 0.598	B7	1.44	1.18	0.87	0.71	0.41	
B8	0.586	0.550	0.556	0.543	0.542	0.589	0.472	0.589	B8	1.50	1.03	0.99	0.62	0.31	
B9	0.581	0.572	0.593	0.546	0.553	0.557	0.404	0.593	B9	1.49	1.21	0.79	0.68	0.39	
B10	0.581	0.588	0.541	0.530	0.519	0.553	0.561	0.588	B10	1.50	1.03	0.71	0.52	0.40	
B11	0.592	0.593	0.593	0.536	0.548	0.548	0.583	0.592	B11	1.46	1.13	1.10	0.66	0.38	
B12	0.572	0.563	0.558	0.526	0.529	0.514	0.471	0.572	B12	1.44	1.20	0.87	0.77	0.41	
B13	0.585	0.586	0.579	0.541	0.537	0.532	0.501	0.586	B13	1.45	1.19	0.88	0.71	0.41	
B14	0.575	0.581	0.570	0.515	0.531	0.529	0.474	0.581	B14	1.49	1.17	0.92	0.70	0.40	
B15	0.570	0.553	0.550	0.516	0.517	0.497	0.432	0.570	B15	1.45	1.19	0.87	0.71	0.41	
B16	0.570	0.548	0.559	0.511	0.520	0.519	0.515	0.570	B16	1.46	1.17	0.88	0.65	0.41	
MAX	0.594	0.598	0.593	0.564	0.569	0.589	0.596	0.598							
MIN	0.570	0.548	0.541	0.511	0.515	0.497	0.404								
STDEV	0.00709	0.016775	0.01842	0.01493	0.015739	0.026481	0.051419								
											Gbest				
											B7	1.44	1.18	0.87	0.71

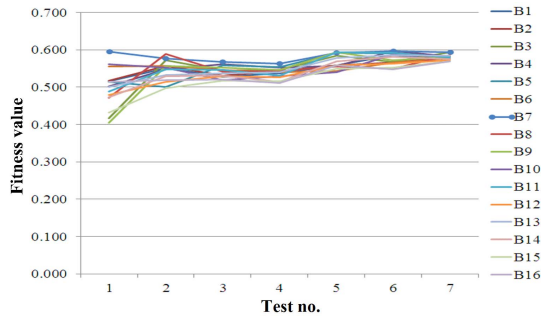


Figure 7. Convergence curve of fitness value.

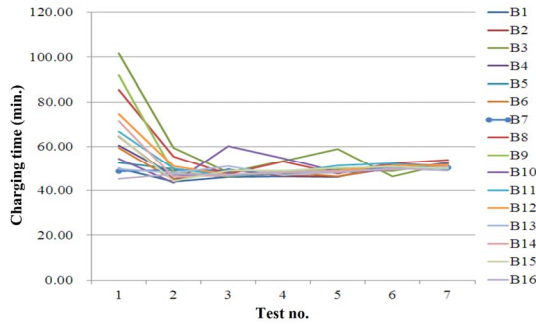


Figure 8. Convergence curve of CT.

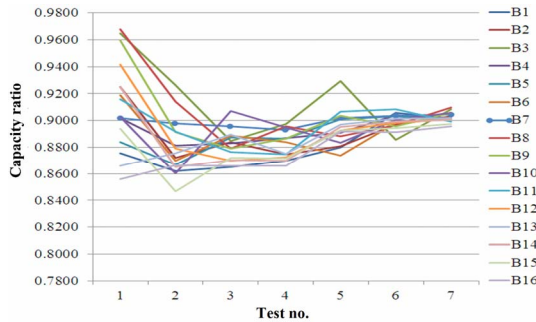


Figure 9. Convergence curve of NDC.

Fig. 10 shows the capacity versus cycles for Li-ion batteries using the obtained charge pattern and the conventional CC-CV method. From Fig. 10, the cycling number of the proposed algorithm when the capacity drop to 88% of the original capacity is 267; the cycling number of the CC-CV method is 221. That is, the proposed charge pattern can provide about 21% more cycle count than the conventional CC-CV method. In other words, the obtained charging pattern is more cost-effective to the Li-ion batteries since it can prolong the cycle life.

IV. CONCLUSIONS

In this paper, a global optimization technique based on the PSO algorithm is introduced to fast charging-pattern searching problems for enhancing the available performance of the Li-ion batteries. Experimental findings show that the resulted pattern is capable of charging the batteries to over 90% available capacity within 50 minutes. Comparing with the conventional constant current–constant voltage (CC-CV) method, the devised scheme has the performance enhancements of more than 80% charging time reduction,

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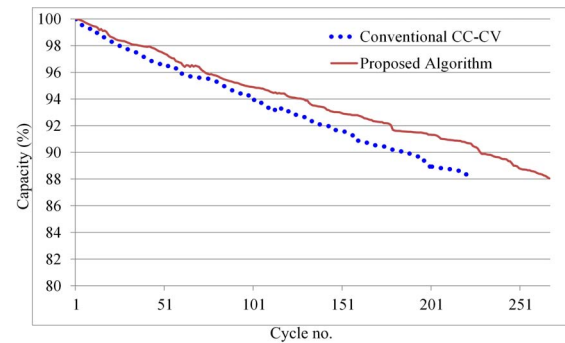


Figure 10. Cycling test of the proposed and conventional CC-CV algorithms.

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